

Abilasha Rameshkumar

Mechanical Engineering • Toronto, ON • abilasharameshkumar@gmail.com • [Linkedin Profile](#) | [Portfolio](#)

EDUCATION

Toronto Metropolitan University

Sept. 2021 - Present

- B.Eng - Mechanical Engineering, Minor in French and Philosophy
- Relevant Courses: Mechanical Design (Capstone), Design for Manufacturing (DFMA), Advanced Manufacturing, Materials Selection, Project Management, Control Systems, Standard Engineering Courses.

SKILLS

Technical Skills: Solidworks, AutoCAD, GD&T, DFMA, MS Suite, MATLAB, Novice Python, CNC/MasterCAM, Dobot, FDM

Professional Skills: Project Coordination, Technical Problem Solving, Team Collaboration, Organization, Attention to Detail

Languages: English (Fluent), Tamil (Fluent), French (Limited Working Proficiency - French Bilingual Certificate)

PROJECTS

Automated Garden Hose Caddy | *Mechanical Capstone Project*

Jan. 2026 - April 2026

- Team leader for a mechanical capstone project developing a competitive, market-ready automated hose management system to eliminate manual handling, improve user ergonomics, and enable controlled hose extension and retraction.
- Designed and modeled the reel subsystem (shaft, drum, flanges, bearings, ratchet and pawl, spring) and integrated full system (reel subsystem, frame, wheels) in Solidworks CAD, ensuring proper alignment, load support and manufacturability of a rotating mechanical system. Produced engineering drawings with GD&T.
- Performed torque, force, and stress-based calculations (spring sizing, bending, shear, and factor of safety) to size the constant-force spring for a 250 ft hose and select Aluminum 6061 for structural components and HDPE for non-structural elements, ensuring sufficient retraction force, structural reliability, and durability under frictional loading and outdoor conditions
- Collaborated with a multidisciplinary team and used Microsoft Project to organize workflows via GANTT charts and track design iterations to meet timelines and product goals.

Tractor Pull Slot Car | *Mechanical Design & Manufacturing Project*

Sept. 2025 - Dec 2025

- Designed and built a load-bearing slot car system capable of climbing a 30° incline under increasing loads, achieving 2nd place.
- Assembled a 3D model in SolidWorks, including the chassis, wheel assembly, and load interface, ensuring proper force transmission and stability. Produced CAD assemblies and engineering drawings.
- Performed traction, friction, and load analysis, iteratively refining the design through prototyping to improve traction and load capacity, resolving wheel slip by selecting high-traction rubber bands and adjusting weight distribution through added mass.
- Manufactured components using CNC (MasterCAM) machining and FDM 3D printing. Delivered a working prototype.

Two-Finger Robotic Gripper | *Mechanism Design Project*

Jan. 2025 - April 2025

- Designed and prototyped a two-finger mechanical gripper capable of reliably grasping and placing 3 lb objects under material and structural constraints.
- Modeled the mechanism in Solidworks, including linkages, joints, and actuation components, and produced engineering drawings for fabrication. Performed force and kinematic analysis to refine model using MATLAB.

WORK EXPERIENCE

Facility in Charge - City of Toronto

July. 2023 - Present

L'Amoreaux Tennis Centre | 300 Silver Springs Blvd, Toronto, ON M1V 1S4

- Coordinating daily operations while ensuring compliance with safety regulations, operational standards, and facility policies.
- Maintaining scheduling, maintenance activities, and program delivery to support efficient and reliable facility operations.
- Maintain structured documentation, incident reports, and communication logs to support smooth facility management.

Private Tutor - Mathematics, Science, & French

Jul. 2021 - Present

- Providing one-on-one academic support to 20+ students in math, science, and French, simplifying complex concepts and improving student performance consistently to a B+ - A+ levels.
- K- 12 in Math, Physics, and French. Tutored courses such as Advanced Functions, Gr 11 & 12 Physics, and Calculus & Vectors.

Technical Support - Co-op

Sept. 2019 - Jan. 2020

Marquis Book Printing | 3480 Pharmacy Ave, Toronto, ON M1W 2S7

- Developed and maintained a centralized Excel-based user and asset database to support system troubleshooting and information management for company users and customers.
- Updated, restored, and standardized branch computer systems to ensure reliable software performance and improve operational productivity.

EXTRA CURRICULARS

President - Tamil Students Association | Toronto Metropolitan University

Sept. 2023 - Aug 2024

- Led a team of 30+ executives, overseeing planning and execution of 30+ events, including multiple large-scale events with 1,000+ attendees.
- Managed budgets, fundraising initiatives, sponsorship coordination, collaboration with other universities, and event logistics, ensuring successful delivery of high-impact community and cultural programs/events.